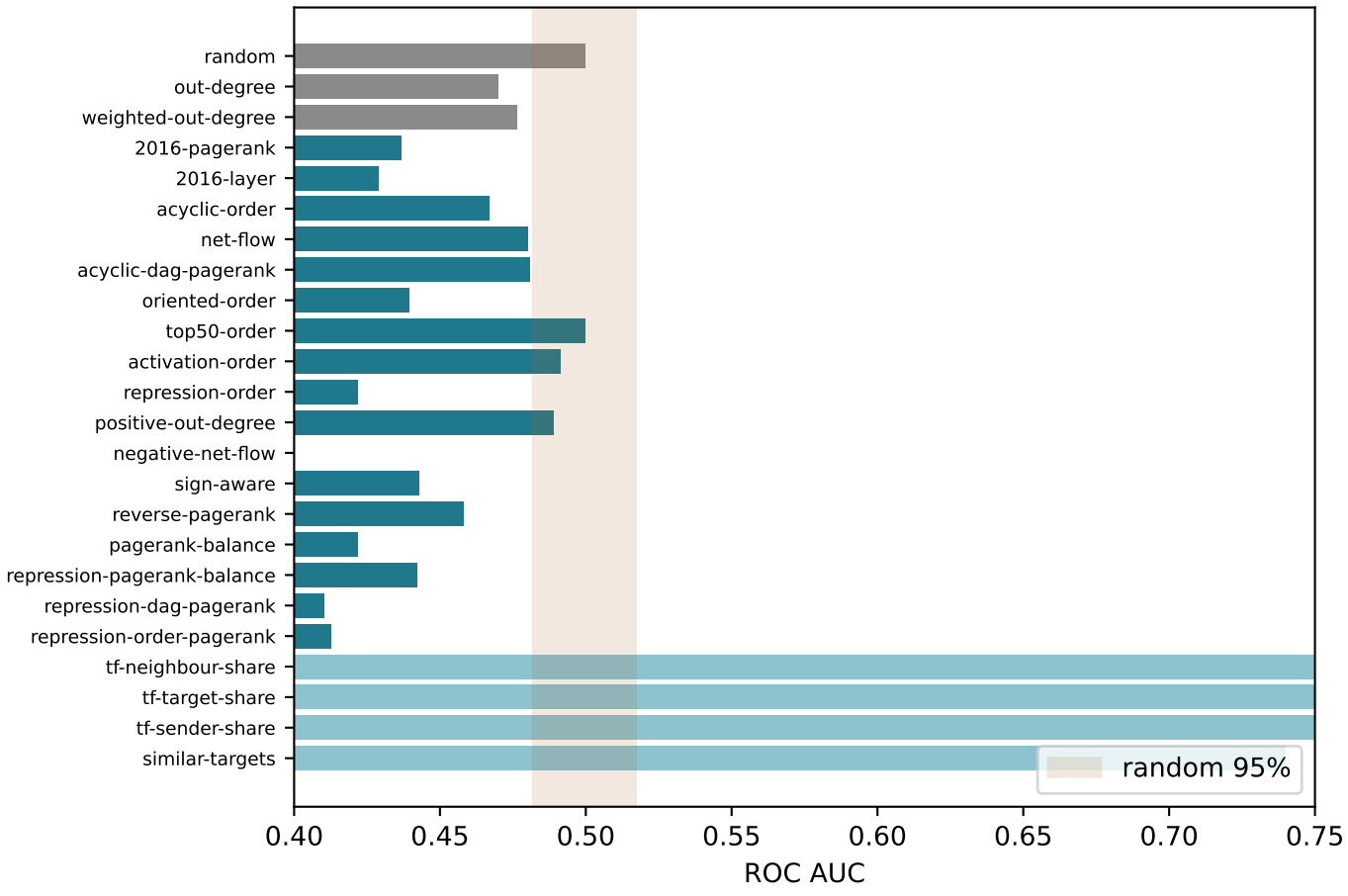
Regulator ranking against reference 'lambert-2018'

2026-11-10 · 11,994 genes ranked, 1,155 in lambert-2018-tfs.txt · top k = 100

Pipeline	Reference genes in top 100	AUC	p top-k	p AUC	AUC vs best baseline (95%)	p vs baseline
random	9.6 (4–16)	0.500 (0.482–0.517)				
out-degree	11	0.470	0.361	0.999	-0.006 (-0.008, -0.005)	< 0.001
weighted-out-degree	10	0.476	0.486	0.993	+0.000 (+0.000, +0.000)	
2016-pagerank	3	0.437	0.997	1.000	-0.039 (-0.051, -0.028)	< 0.001
2016-layer	14	0.429	0.102	1.000	-0.047 (-0.062, -0.033)	< 0.001
acyclic-order	13	0.467	0.160	0.999	-0.009 (-0.016, -0.002)	0.016
net-flow	13	0.480	0.160	0.984	+0.004 (-0.005, +0.014)	0.390
acyclic-dag-pagerank	14	0.481	0.102	0.982	+0.005 (-0.001, +0.010)	0.108
oriented-order	6	0.439	0.927	1.000	-0.037 (-0.047, -0.027)	< 0.001
top50-order	10	0.500	0.486	0.511	+0.024 (+0.006, +0.041)	0.010
activation-order	11	0.491	0.361	0.842	+0.015 (+0.007, +0.023)	< 0.001
repression-order	7	0.422	0.864	1.000	-0.054 (-0.066, -0.043)	< 0.001
positive-out-degree †	7	0.489	0.864	0.889	+0.013 (+0.010, +0.016)	< 0.001
negative-net-flow †	6	0.400	0.927	1.000	-0.076 (-0.088, -0.063)	< 0.001
sign-aware †	12	0.443	0.239	1.000	-0.033 (-0.040, -0.027)	< 0.001
reverse-pagerank †	9	0.458	0.621	1.000	-0.018 (-0.021, -0.015)	< 0.001
pagerank-balance †	8	0.422	0.754	1.000	-0.055 (-0.066, -0.043)	< 0.001
repression-pagerank-balance †	10	0.442	0.486	1.000	-0.034 (-0.046, -0.022)	< 0.001
repression-dag-pagerank †	8	0.410	0.754	1.000	-0.066 (-0.078, -0.052)	< 0.001
repression-order-pagerank †	7	0.413	0.864	1.000	-0.064 (-0.076, -0.052)	< 0.001
tf-neighbour-share ‡	99	0.780	< 0.001	< 0.001	+0.303 (+0.280, +0.327)	< 0.001
tf-target-share ‡	97	0.764	< 0.001	< 0.001	+0.288 (+0.263, +0.311)	< 0.001
tf-sender-share ‡	99	0.771	< 0.001	< 0.001	+0.294 (+0.270, +0.317)	< 0.001
similar-targets ‡	91	0.740	< 0.001	< 0.001	+0.264 (+0.237, +0.286)	< 0.001

Random: mean and 95% range over random orderings. p: share of random orderings scoring at least as well. Grey bars: random and baselines. AUC vs best baseline: paired bootstrap over genes (1,000 resamples). †: designed after seeing these results; needs confirming on other data. ‡: uses known TFs: scored from other genes' labels in this reference (lighter bars).



Edge checks against curated TF→target pairs

Recall ×: share of curated pairs present as edges, over chance.
Positive/negative ×: share of edges of that sign that are curated pairs, over chance. Sign: agreement with the curated sign (chance).

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Graph	Pairs	Recall (×)	Positive ×	Negative ×	Sign (chance)
raw graph	31,067	0.047 (2.73×)	3.34×	0.88×	0.81 (0.80)
resolve_reciprocals → drop_sinks → prune_triangles	31,042	0.005 (1.13×)	3.06×	0.91×	0.40 (0.34)
resolve_reciprocals → drop_sinks → prune_triangles → greedy_layering	23,059	0.003 (0.97×)	2.23×	0.92×	0.26 (0.23)
absolute	31,067	0.047 (2.73×)			
absolute → minimum_feedback_order	31,011	0.033 (2.47×)			
orient_reciprocals ratio=2.0 → absolute	30,981	0.018 (1.85×)			
orient_reciprocals ratio=2.0 → absolute → minimum_feedback_order	30,981	0.018 (1.85×)			
top_outgoing k=50 → absolute	31,067	0.017 (4.48×)			
top_outgoing k=50 → absolute → minimum_feedback_order	30,939	0.011 (4.06×)			
keep_sign sign=1 → absolute	31,067	0.044 (3.34×)			
keep_sign sign=1 → absolute → minimum_feedback_order	31,019	0.030 (3.08×)			
keep_sign sign=-1 → absolute	29,687	0.004 (0.86×)			
keep_sign sign=-1 → absolute → minimum_feedback_order	28,396	0.004 (0.94×)			
keep_sign sign=-1	29,687	0.004 (0.86×)		0.86×	0.23 (0.23)

Pipeline steps

out-degree: Most outgoing edges (raw graph)

score out_degree, descending

Graph: 2,620,245 edges (628,004 negative), largest strong component 11,994.

weighted-out-degree: Largest total outgoing weight (raw graph)

score weighted_out_degree, descending

Graph: 2,620,245 edges (628,004 negative), largest strong component 11,994.

2016-pagerank: 2016 cleaning, lowest PageRank

resolve_reciprocals() → drop_sinks() → prune_triangles() → score pagerank, ascending

Graph: 606,144 edges (547,820 negative), largest strong component 11,549.

2016-layer: 2016 cleaning, earliest greedy layer

resolve_reciprocals() → drop_sinks() → prune_triangles() → greedy_layering() → score level, ascending

Graph: 606,144 edges (547,820 negative), largest strong component 11,549; DAG keeps 365,272 edges.

acyclic-order: Earliest in acyclic order

absolute() → minimum_feedback_order() → score level, ascending

Graph: 2,620,245 edges (0 negative), largest strong component 11,994; DAG keeps 1,962,874 edges.

net-flow: Outgoing minus incoming weight

absolute() → score net_flow, descending

Graph: 2,620,245 edges (0 negative), largest strong component 11,994.

acyclic-dag-pagerank: Reverse PageRank on the acyclic DAG

absolute() → minimum_feedback_order() → score dag_reverse_pagerank, descending

Graph: 2,620,245 edges (0 negative), largest strong component 11,994; DAG keeps 1,962,874 edges.

oriented-order: Pairs with a 2x stronger side only, earliest in acyclic order

orient_reciprocals(ratio=2.0) → absolute() → minimum_feedback_order() → score level, ascending

Graph: 1,399,036 edges (0 negative), largest strong component 10,471; DAG keeps 1,382,204 edges.

top50-order: Top 50 outgoing edges per gene, earliest in acyclic order

top_outgoing(k=50) → absolute() → minimum_feedback_order() → score level, ascending

Graph: 559,454 edges (0 negative), largest strong component 11,986; DAG keeps 388,639 edges.

activation-order: Activating edges only, earliest in acyclic order

keep_sign(sign=1) → absolute() → minimum_feedback_order() → score level, ascending

Graph: 1,992,241 edges (0 negative), largest strong component 11,990; DAG keeps 1,430,491 edges.

repression-order: Repressing edges only, earliest in acyclic order

keep_sign(sign=-1) → absolute() → minimum_feedback_order() → score level, ascending

Graph: 628,004 edges (0 negative), largest strong component 10,673; DAG keeps 540,157 edges.

positive-out-degree: Most positive outgoing edges

score positive_out_degree, descending

Graph: 2,620,245 edges (628,004 negative), largest strong component 11,994.

negative-net-flow: Negative outgoing minus incoming weight

score negative_net_flow, descending

Graph: 2,620,245 edges (628,004 negative), largest strong component 11,994.

sign-aware: Rank sum: positive out-degree and negative net flow

score sign_aware, descending

Graph: 2,620,245 edges (628,004 negative), largest strong component 11,994.

reverse-pagerank: Highest reverse PageRank (raw graph)